

Programming for Business Analytics Coursework

Word count - 1496

Introduction

In today's global economy, understanding how key macroeconomic indicators behave across countries is essential for anyone involved in financial markets. Investors, governments, and institutions rely on data like GDP, inflation, interest rates, and employment to make informed decisions. This project aims to explore how these indicators change over time and how they relate to one another, helping us understand broader economic trends and risks.

The research question - "How do changes in macroeconomic indicators such as inflation, interest rates, and GDP per capita relate to one another across different countries, and what insights can these relationships provide for financial market stakeholders?" - aims to uncover meaningful insights from real-world economic data by analysing relationships between key indicators across multiple countries. This will help us understand how different economies respond to global events, shifts in investor confidence, and changing market conditions.

The project focuses on the period from 2000 to 2023, a timeline chosen to capture major global events including the Dot-Com Bubble, the 2008 Financial Crisis, the Eurozone Debt Crisis and COVID-19. Analysing this time frame allows us to see how economies react to shocks and how those reactions show up in their data.

What makes this project stand out is its global, comparative perspective. Rather than focusing on a single country, we look across nations to spot patterns and highlight indicators that might signal economic risk or recovery. The goal is to turn raw economic data into useful insights for different market stakeholders through data cleansing and visualisation techniques.

Data Collection

Initially, I explored websites like Kaggle and Reddit in search of suitable datasets. While some were interesting, covering topics like stock indices and job satisfaction, they often lacked the required time range (2000–2023) or had too few rows to meet the project's requirements. Many datasets were also based on inconsistent timeframes, which would have made it difficult to compare indicators across countries or track trends through major global events. Combining data from multiple sources also caused problems, especially due to mismatched dates and missing values. This would have made comparisons unreliable and graphs harder to interpret, particularly when looking at long-term economic patterns.

To overcome this, I turned to World Bank Open Data, which offered a reliable, comprehensive source of economic indicators. Their datasets were large enough, consistent in format, and covered the full time period I needed. More importantly, they provided a wide range of global economic indicators, allowing me to select metrics that reflected different aspects of an economy such as inflation, GDP per capita, labour force participation, and interest rates.

I specifically chose indicators that would offer a well-rounded view of economic performance across countries. This ensured that I could compare nations not just on overall growth, but also

on structural strengths and weaknesses, highlighting where a country might excel in one area while struggling in another.

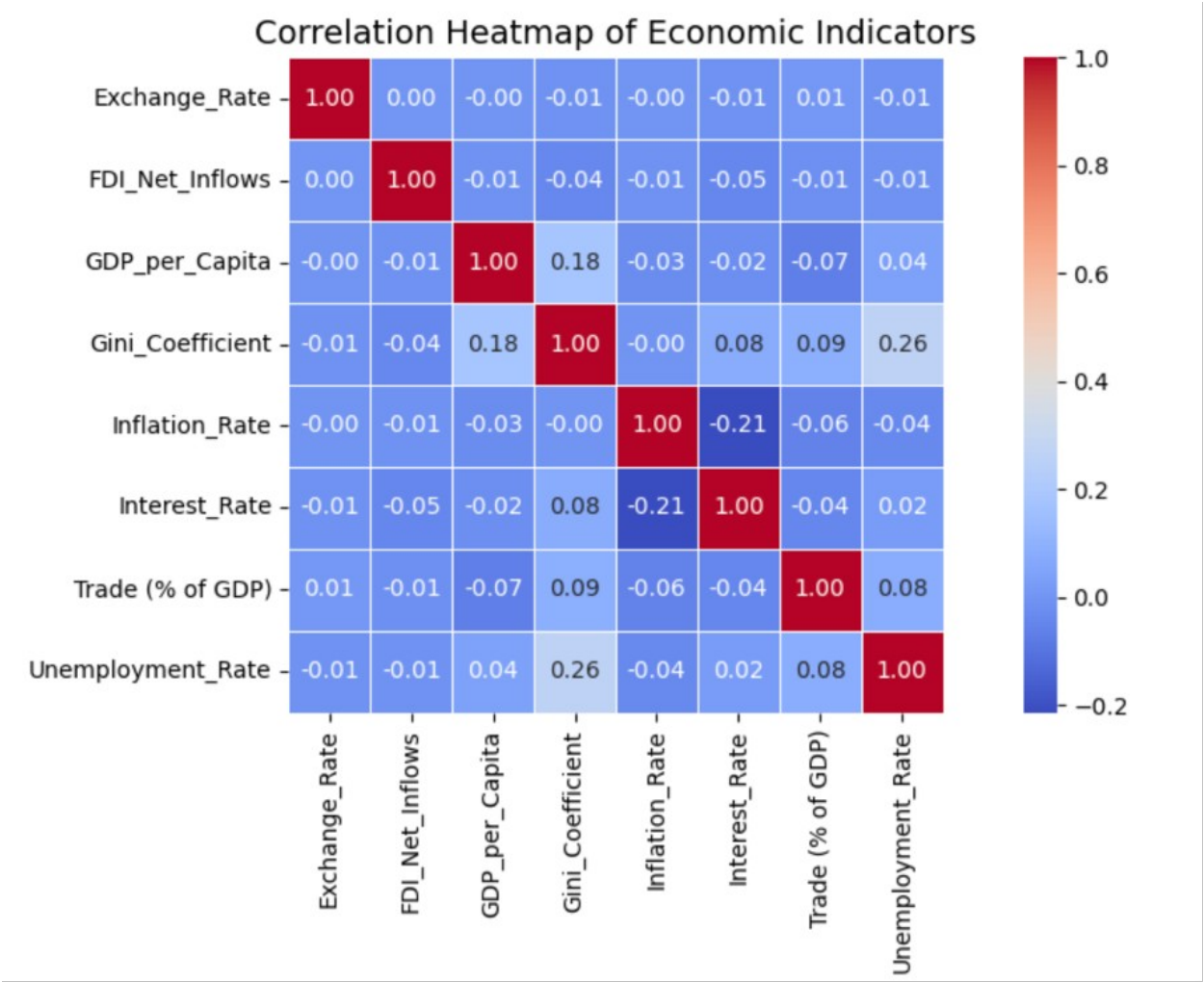
Data Cleansing

I began the data cleaning process by renaming the column headers to more appropriate names, making the dataset easier to navigate and understand as well as following better coding practices. The dataset contained missing values which I had to manually convert to NaNs, which needed to be handled before analysis. Initially, I experimented with filling missing values using means, but eventually found that replacing NaNs with the median produced more stable and reliable results, as the median is less sensitive to extreme values. Before calculating medians, I filled the missing values with zeros—this ensured the median calculation wouldn't ignore NaNs entirely, which could skew results.

Once cleaned, I reformatted the dataset using a pivot operation, converting it from long to wide format with the help of Stack Overflow (2015). This allowed each economic indicator to become a separate column, making it easier to compare values across countries and years. Restructuring the dataset in this way significantly improved the efficiency of analysis and visualization, as I could easily isolate specific indicators for plotting and comparison. Without pivoting, the indicators were stacked in a single column, which made comparisons and graphing far more complicated. The dataset was structured with each row representing a unique country-year combination, and each economic indicator as its own column. This layout enabled both cross-sectional (country-to-country) and longitudinal (year-over-year) analysis.

Zabihullah¹⁸ (2025) provided valuable insights that influenced the data cleaning process in this project. Their approach offered practical techniques for handling missing values and improving the structure and presentation of the dataset. Additionally, their work helped inform how economic insights could be tailored to different market participants.

Data Analysis



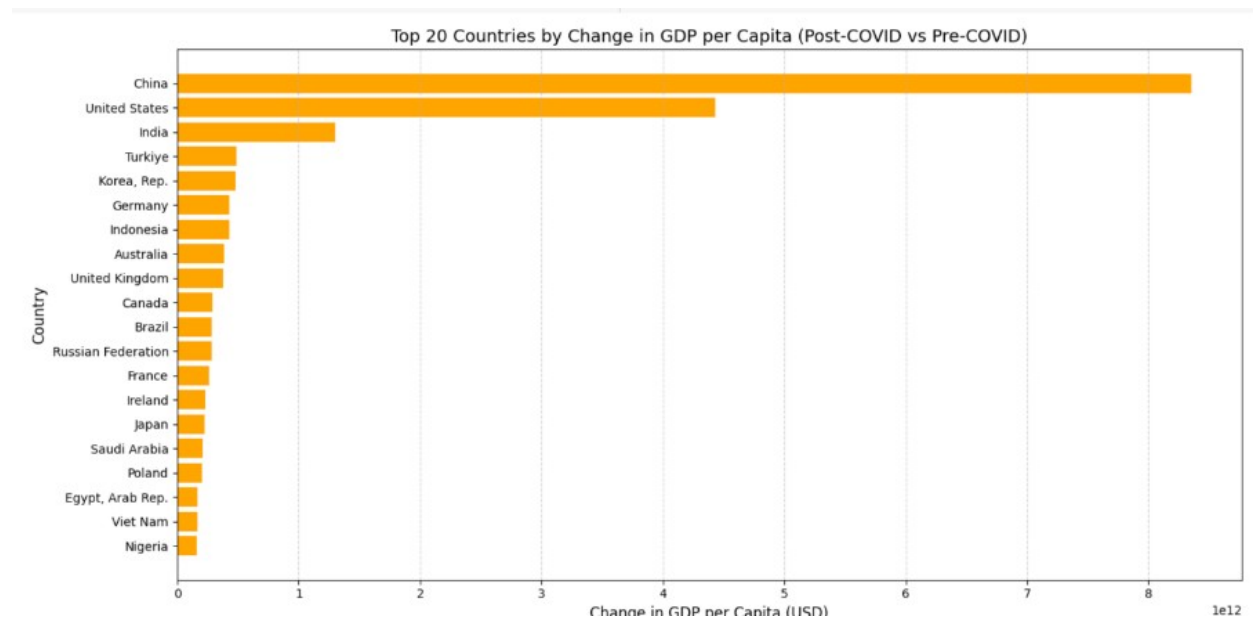
The heatmap inspired by (W3schools, 2024) illustrates how different economic indicators relate to one another across countries and time, using Pearson correlation values ranging from -1 (strong negative correlation) to +1 (strong positive correlation).

The most notable pattern is a weak to moderate positive correlation (+0.26) between income inequality (Gini coefficient) and unemployment. This suggests that higher inequality may be linked to higher unemployment, possibly due to reduced access to education or job opportunities among lower-income populations.

A slight negative correlation (-0.21) between interest and inflation rates also appears, aligning with how central banks raise interest rates to control inflation. Interestingly, GDP per capita shows very weak correlations with other indicators . Only +0.18 with Gini and +0.04 with unemployment suggesting that wealth alone doesn’t strongly predict inequality or labor market conditions on a global scale.

Other indicators, such as exchange rate, FDI net inflows, and trade (% of GDP), show little to no correlation with most variables.

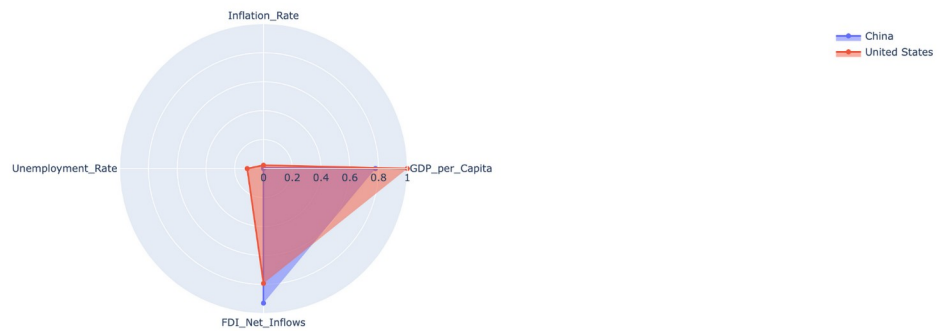
In summary, the heatmap reveals that global economic indicators generally have weak linear relationships with each other. This highlights the complexity of global economies and reinforces the need for more targeted, country-level analysis.



The bar chart highlights how GDP per capita shifted between the pre- and post-COVID periods among the top 20 countries. China recorded the most significant rise, likely due to its swift pandemic response, early industrial rebound, and strong export momentum. The U.S. also saw a large increase, driven by fiscal stimulus and a booming tech sector, although inflation may have played a role. India followed closely, reflecting rapid growth in its services sector and strong domestic demand. In contrast, developed economies like Germany, Japan, and the UK showed smaller gains, possibly due to slower reopenings, supply chain issues, or tighter fiscal policy. Emerging economies such as Indonesia, Vietnam, Nigeria, and Egypt also appeared on the list, albeit with lower absolute increases, suggesting a broader but uneven global recovery. From a financial markets lens, strong GDP rebounds can boost investor confidence, attract foreign investment, and stimulate equity markets. Overall, the chart reflects a shifting global economic landscape, where both advanced and emerging nations are playing key roles in the post-crisis recovery.

Mulla, A. (2024) served as a useful reference when developing the graph displaying the top 20 countries. His use of visualisations, economic indicators, and overall report structure provided inspiration for how to approach both the analysis and presentation of data in this project. The notebook also offered ideas on how different indicators could be used together to extract meaningful insights through effective data visualisation.

Economic Profile Comparison (Radar Chart)



The radar chart which I created with the aid of Plotly (2024) compares key economic indicators such as GDP per capita, inflation, unemployment, and FDI net inflows for China and the United States. As expected, the U.S. leads in GDP per capita, reflecting its status as a mature, high-income economy, while China trails behind but remains on a strong growth trajectory. Interestingly, China surpasses the U.S. in FDI net inflows, likely due to its dominance in manufacturing and attractive economic zones. The U.S. shows slightly higher unemployment, which may be tied to more transparent reporting or recent economic shocks. Both countries report similarly low inflation at the time of measurement. Overall, the chart reveals key structural differences: while the U.S. excels in income and wealth, China stands out in investment appeal and labour market consistency. This multi-indicator comparison gives a clear economic snapshot, useful for both policymakers and investors assessing global competitiveness and strategic opportunities.

Conclusions and recommendations

Over the course of this project, I explored key macroeconomic indicators across more than 200 countries from 2000 to 2023, using data science tools to uncover trends in GDP per capita, inflation, interest rates, unemployment, and more. By cleaning and reshaping the data, then applying visual analysis, I was able to draw out meaningful relationships and differences

between countries, especially in how they responded to major events like the 2008 financial crisis and the COVID-19 pandemic.

While some indicators like interest and inflation rates behaved in line with economic theory, others had weaker or more complex relationships, reminding us that each country's economy is shaped by its own unique factors.

Overall, this project shows that with the right tools and approach, raw economic data can be turned into something practical and meaningful. Whether you're making policy decisions, investment choices, or planning international support, these insights can help you understand not just where a country stands today—but where it might be headed tomorrow.

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Appendix

https://drive.google.com/drive/folders/1P5g9GOknlzBVBmfolnxUNc0_ybDZkegD?usp=sharing